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July 9, 1968

INTERIM FEDERAL SPECIFICATION

FLASHLIGHT (PENLIGHT)

This Interim Federal Specification was developed by General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20405, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers small pocket size dry cell flashlights of tubular design with pocket clip.

1.2 Classification.

1.2.1 Types. The flashlight furnished under this specification shall be of the following types (see 6.1):

Type I - With translucent shield.
Type II - With opaque shield.

2.1 The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.:

Federal Specifications:

W-B-101 - Battery, Dry.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Std. No. 123 Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is on sale on the subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 6230

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The flashlight furnished under this specification shall be of tubular construction consisting of cap, case, lamp, lampholder, switch, and clip. Batteries shall not be furnished with the flashlight.

3.2 Material. The materials used throughout shall be suitable in every respect for their intended purpose and such that the finished components shall evidence no indication of damage, protective coat penetration, or corrosion when the flashlight is subjected to the tests specified in section 4. Dissimilar metals shall not be used in intimate contact with each other unless protected against electrolytic action.

3.2.1 Plastic. Plastic used in the manufacture of flashlights shall not deteriorate or become harmful to personnel when exposed to water, heat, oil, or acid fumes and shall meet all requirements of this specification.

3.3 Construction.

3.3.1 The flashlight shall be so constructed that no parts (excluding threaded parts) will work loose due to normal lifetime usage, shipping, or when tested in accordance with section 4.

3.3.2 Interchangeability. All corresponding parts of flashlights of the same type and manufacture shall be mechanically and electrically interchangeable.

3.3.3 Housing. The battery housing shall be of chromeplated, copper coated steel, brass, aluminum, or of a platic that meets all applicable requirements and tests specified in this specification. The flashlight shall be designed to accommodate standard AA dry cells which are in accordance with W-B-101.

3.3.4 Lamp head and shield.

3.3.4.1 Lamp head. The lamp head shall include a lamp shield, lamp socket, and a miniature screw base lamp. The lamp socket shall be designed to solidly retain the lamp in position by one or more threads. The threads shall not strip or ride over mating parts when a torque of 20 inch-pounds, plus or minus 1 inch-pound is used.

3.3.4.2 Shield. The head shall be provided with a translucent (type I) or opaque (type II) shield or housing that shall cover the lamp. The end of the shield or housing shall be provided with an opening of 0.25 inches, plus or minus 0.08 inches, for the projection of light.

3.3.5 Threads. Unless otherwise specified, all threads shall be designed not to strip or override mating parts when a torque of 20 inch-pounds, plus or minus 1 inch-pound is applied. The thread component parts shall be designed to screw up tight when a maximum torque of 7 inch-pounds, plus or minus 1 inch-pound, is applied.

3.3.6 Switch. The flashlight shall be provided with a suitable "off-on" switch that will be solidly retained in position, when placed in either the "on" or "off" positions. The switch shall be designed to enable the operator, with one hand, to easily hold and actuate the switch. The design and the construction of the switch shall be such that it will operate 25,000 cycles at a minimum rate of 15 cycles per minute and not to exceed 30 cycles per minute.

3.3.7 Pocket clip. The pocket clip shall be designed to firmly hold the flashlight in a shirt pocket. The clip shall be located on the body of the flashlight, near the end opposite to the lamp head. The clip shall be designed so that the switch cannot be accidently energized when the flashlight is clipped in the pocket.

3.3.8 Dimensions. The overall dimensions of the flashlight shall not exceed 5.75 inches long and 0.750 inch body diameter.

3.3.9 Cleaning. All parts shall be thoroughly cleaned. Traces of corrosive compounds used for cleaning, plating, finishing, or machining shall be removed or chemically neutralized to prevent subsequent corrosion.

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3.3.10 Workmanship. The flashlight shall be fabricated and finished in a thoroughly workmanship-like manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; marking and dimensions; thoroughness of soldering, painting, and riveting, alignment of parts and tightness of assembly screws.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be made at room temperature, pressure, and humidity.

4.2 Sampling.

4.2.1 Visual and mechanical. Sampling for visual and mechanical inspection shall be conducted in accordance with MIL-STD-105, inspection level S4, acceptable quality level (AQL) 2.5.

4.2.2 Tests. Sampling for test shall be conducted in accordance with MIL-STD-105, inspection level S3, (AQL) 2.5.

4.3 Inspection. A visual and mechanical inspection shall be conducted in accordance with, but not limited to, the following classification of defects.

Visual and Mechanical Inspection

Materials, parts, or accessories, missing or not as specified.
Dimensions not as specified.
Threaded parts not as specified.

4.4 Test.

4.4.1 High temperature. One half of the flashlights in a sample, completely assembled including batteries, shall be subjected to a temperature of 140°F plus or minus 2°F. dry heat for 16 hours followed by a temperature of 140°F plus or minus 2°F at 90 to 100 percent relative humidity for 6 hours. The flashlight shall then be compared with untested flashlights for operation, dimensional stability, crazing of surface, changes in appearance of materials, or any other defects. The parts of the tested samples shall be interchangeable with corresponding parts of untested samples and after new batteries have been installed the test samples shall be operational.

4.4.2 Plastic. The plastic parts from 3 flashlights shall be submerged in mineral oil at a temperature of 100°F plus or minus 2°F for a period of 15 minutes. The component parts shall then be wiped clean and submerged in a 10 percent solution of sodium chloride at a temperature of 80°F plus or minus 5°F for a period of 30 minutes. Upon completion of the above test the flashlights shall be examined carefully and evidence no indication of crazing, cracking, dimensional change, or any chemical change that can be detected with normal vision and physical examination. The tested parts shall be interchanged with like parts of untested flashlights. The failure of any component part to pass this test will constitute failue of this test (see 3.2.1).

4.4.3 Threads. Starting with the case of the flashlight disassembled, it shall be assembled, applying a measured torque of 20 inch-pounds, plus or minus 1 inch-pound, to all threaded parts of the case. The flashlight shall then be examined to determine compliance with requirement 3.3.5. The flashlight shall be disassembled again and reassembled applying a torque of 7 inch-pounds, plus or minus 1 inch-pound, to all threads. The flashlight shall then be examined for compliance with the stated requirement of 3.3.5.

4.4.4 Switch. The sustained contact switch mechanism of the flashlight shall be tested by operating the switch from full "off" position to full "on" position and back to "off" position, which shall be considered as one cycle. The switch shall be operated under normal electrical load conditions throughout the required 25,000 cycles of operation. Upon completion of the above test the flashlight shall function properly and the light shall not go off when shaken vigorously with the switch in the "on" position. The test shall be conducted in accordance with 3.3.6.

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4.4.5 Pocket clip. The pressure point of the pocket clip shall be pulled away from the housing a minimum distance of 0.125 inches for a total of 100 times. A piece of metal with rounded corners may be used for this test. Upon completion of the above test a sheet of paper 0.005 inches thick, or less, shall be inserted into the clip past the pressure point. With the penlight in a stationary position it must take more than a two ounce weight to pull the paper clear of the clip. There shall be no distortion, breakage or reduction of holding pressure throughout this test.

4.4.6 Impact. The flashlight shall be subjected to a temperature of minus 30°F, plus or minus 2°F, for two (2) hours and then shall be dropped in a free fall throughout a distance of 40 inches onto a hardwood floor. The flashlight shall be dropped so that the longitudinal axis of the case will be parallel to the floor. Failure of the flashlight to operate normally after the test, or any chipping, cracking, shall constitute failure of this test.

4.4.7 Drop. The assembled flashlight, complete with dry cells, lamp, shield, and case shall be dropped in free fall through a distance of 40 inches onto a hardwood floor. The flashlight shall be dropped 3 times on the side of the case and once on the lamp end. There shall be no crazing, or any other defects which interfere with the proper operation of the flashlight. Upon the completion of this test the flashlight shall function properly and the light shall not go off when shaken vigorously with the switch in the "on" position.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 Level A. Each flashlight shall be packaged in accordance with MIL-P-116, submethod IC-2.

5.1.2 Level B. Each flashlight shall be packaged in a snug-fitting paperboard box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. Each box shall be secured to prevent accidental opening.

5.1.3 Level C. The flashlight shall be packaged to afford adequate protection against deterioration and damage during shipment from the supplier to the initial destination.

5.1.4 Intermediate packaging. Unless otherwise specified (see 6.1), twenty-four flashlights shall be packaged in a snug-fitting paperboard box conforming to PPP-B-566, or PPP-B-665, or PPP-B-676. The box shall be secured to prevent accidental opening.

5.2 Packing.

5.2.1 Level A. Flashlights of one size only in quantities as specified (see 6.1), shall be packed in a snug-fitting fiberboard box conforming to PPP-B-636, class weather-resistant, grade V3c. The gross weight shall not exceed the weight limitations of the box specification.

5.2.2 Level B. Twenty intermediate packages (480 flashlights) shall be packed in a snug-fitting fiberboard box conforming to PPP-B-636, class domestic, grade 275. Closure shall be in accordance with the appendix to PPP-B-636.

5.2.3 Level C. The flashlights, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the Uniform Freight Classification Rules or the National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and the shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options offered herein and include the following data in procurement documents:

a. Title, symbol, and date of this specification.
b. Type desired (see 1.2.1).
c. Responsibility for inspection, if other than specified (see 4.1).
d. Selection of applicable level of packing and packaging (see 5.1 and 5.2).
e. Marking (see 5.3).
f. Intermediate packaging if other than specified (see 5.1.4).
g. Quantity required for level A packing (see 5.2.1).

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